

Part B Chapter 15

Driving Behaviour¹

1 Introduction

Recent developments within Ambient Intelligence provide new technological possibilities to contribute to personal care for safety, health, performance, and wellbeing; cf. (Aarts, Harwig, and Schuurmans, 2001; Riva et al., 2005). Applications make use of possibilities to acquire sensor information about humans and their functioning, and knowledge for analysis of such information. Based on this, ambient devices can (re)act by undertaking actions in a knowledgeable manner that improve the human's, safety, health, performance, and wellbeing.

The focus of this chapter is on driving behaviour. Circumstances may occur in which a person's internal state is affected in such a way that driving is no longer safe. For example, when a person has taken drugs, either prescribed by a medical professional, or by own initiative, the driving behaviour may be impaired. For the case of alcohol, specific tests are possible to estimate the alcohol level in the blood, but for many other drugs such tests are not available. Moreover, a bad driver state may have other causes, such as highly emotional events, or being sleepy. Therefore assessment of the driver's state by monitoring the driving behaviour itself and analysing the monitoring results is a wider applicable option. In this chapter, an ambient agent is presented that assesses a person's driving behaviour, and in case of a negative assessment lets cruise control slow down and stop the car (see Bosse, Hoogendoorn, Klein, and Treur, 2008). The approach was inspired by a system that is currently under development by Toyota. This ambient system that in the near future will be incorporated as a safety support system in Toyota's prestigious Lexus line, uses sensors that can detect the driver's steering operations, and sensors that can detect the driver's eye movements.



¹ Based on: Bosse, T., Hoogendoorn, M., Klein, M., and Treur, J. (2008), A Component-Based Ambient Agent Model for Assessment of Driving Behaviour. In: F.E. Sandnes et al. (eds), *Proceedings of the 5th International Conference on Ubiquitous Intelligence and Computing, UIC'08*. Lecture Notes in Computer Science, vol. 5061. Springer Verlag, 2008, pp. 229-243.

The presented ambient agent makes use of three types of sensor information. First, the driver's steering behaviour is measured, to assess, among others, whether (s)he is making too many unnecessary movements with the steering wheel. Second, a sensor in the steering wheel measures the driver's perspiration. This can be used to assess the percentage of alcohol in the driver's sweat, which correlates linearly with the alcohol percentage in the blood. Third, a camera that is mounted on the front mirror measures the driver's eye movements, which can be used to assess, for instance, whether (s)he is staring too much in the distance, or whether (s)he is blinking much with the eyes.

2 Domain Model

This section introduces the domain model for driver behaviour. In Section 2.1, the main aspects of the model and their relations are introduced. In Section 2.2, the detailed model is provided, and Section 2.3 presents an example simulation trace.

2.1 Analysis of the Main Aspects and Their Relations

In the literature, various factors are reported that have a (negative) impact on driver behaviour. These factors include high levels of drugs or alcohol, fatigue, specific emotional states, the presence of other persons in the car, loud music, and the use of mobile phones. The system described in this chapter is designed for measuring three of these factors. However, it can be extended to account for other factors as well.

First, the *level of drugs* in the blood is considered. This is dependent on the amount of *drug intake* over the last hours. Second, and similarly, the *level of alcohol* in the blood (and also in the sweat) is dependent on the amount of *alcohol intake* over the last hours. The third factor is the person's state of *fatigue*, which depends on whether long periods of *non-stop driving* have been performed recently. All of these factors may lead to a state that negatively influences the driving behaviour of the person. In particular, in case a person has taken certain drugs or alcohol, his or her *steering behaviour* may be affected: (s)he will make more sudden movements in the steering wheel, instead of keeping it straight. In addition, the person's eye movements may become abnormal. A person may start to *stare* (i.e., fixating the eyes on one location, without making any saccades) in case (s)he has consumed a lot of alcohol and/or is very tired. In case of the latter, the person's eyes will also *blink* more often than usually. This happens because the state of fatigue makes that the *speed* increases by which the eyes become *dry*, and also because the *basic eye dryness* (the dryness just after a blink) increases. When a person's eyes become too dry (i.e., they are drier than a certain desired *norm*), a person blinks (i.e., rapidly opens and closes the eyelid) in order to spread tears across and remove irritants².

² For a more detailed description of the relation between eye movements and fatigue, see e.g., (Caffier, Erdmann, and Ullsperger, 2003).

In sum, the following concepts are needed to describe the dynamics of a person's driving behaviour:

- drug intake
- alcohol intake
- non-stop driving
- drug level (in blood)
- alcohol level (in blood and sweat)
- fatigue
- steering behaviour
- staring
- eye blinks
- eye dryness
- basic eye dryness
- eye drying speed
- eye dryness norm

Based on these concepts, the following dynamic relationships can be identified:

Affecting impaired states

- drug intake affects drug level
- alcohol intake affects alcohol level
- non-stop driving affects fatigue

Affecting steering behaviour

- drug level affects steering behaviour
- alcohol level affects steering behaviour

Affecting staring

- alcohol level affects staring
- fatigue affects staring

Affecting basic eye dryness and drying speed

- fatigue affects basic eye dryness
- fatigue affects eye drying speed

Affecting eye dryness

- eye blinks affect eye dryness
- basic eye dryness affects eye dryness
- eye drying speed affects eye dryness

Affecting eye blinks

- eye dryness affects eye blinks
- eye dryness norm affects eye blinks

These dynamic relationships are visualised in Figure 1.

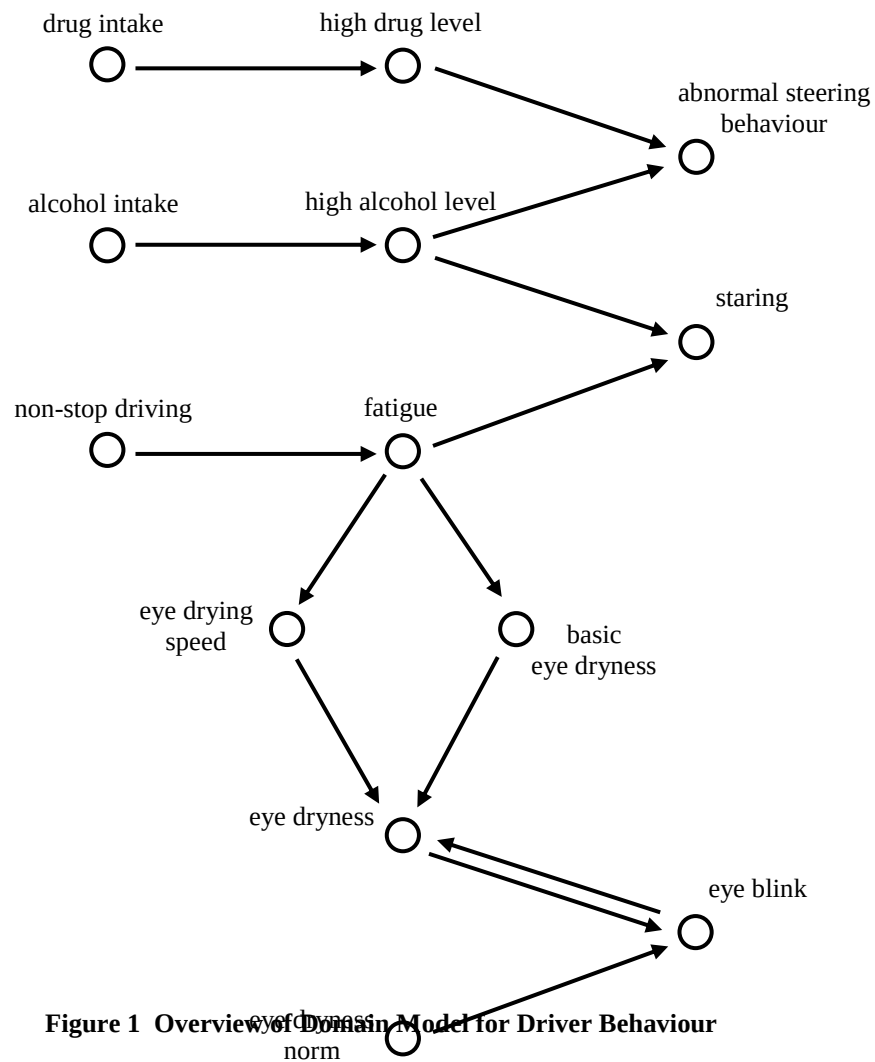


Figure 1 Overview of Domain Model for Driver Behaviour

2.2 The Detailed Model

To formalise the concepts introduced in the previous section, for each of them a logical atom is introduced; see Table 1. The predicate `affects(P:PRODUCT, S:STATE)` is not needed per se, but has been introduced in order to create some more generic relationships.

<i>concept</i>	<i>formalisation</i>
drug intake	<code>consumed(take_drugs)</code>
alcohol intake	<code>consumed(take_alcohol)</code>
non-stop driving	<code>consumed(non_stop_driving)</code>
drug level	<code>has_level(drug_level, L:LEVEL)</code>
alcohol level	<code>has_level(alcohol_level, L:LEVEL)</code>
fatigue	<code>has_level(fatigue, L:LEVEL)</code>
abnormal steering behaviour	<code>performed(abnormal_steering_behaviour)</code>
staring	<code>performed(staring)</code>
eye blinks	<code>performed(eye_blink)</code>
eye dryness	<code>eye_dryness(I:REAL)</code>
basic eye dryness	<code>basic_eye_dryness(I:REAL)</code>
eye drying speed	<code>eye_drying_speed(I:REAL)</code>

eye dryness norm	eye_dryness_norm(I:REAL)
relations between factors and driver states	affects(P:PRODUCT, S:STATE)

Table 1 Formalisation of concepts used for domain model

Note that some atoms make use of sorts. The specific sorts that are used in the presented model, and the elements that they contain, are shown in Table 2. Note that, instead of using the qualitative sort LEVEL, also a quantitative sort can be used (e.g., describing alcohol concentration by a real number).

<i>sort</i>	<i>description of use</i>	<i>elements</i>
LEVEL	extent to which a certain factor (e.g., drugs, alcohol, or fatigue) is present in a person's body	{low, high}
REAL	the set of real numbers	the set of real numbers
PRODUCT	different products that can be consumed	{drugs, alcohol, non_stop_driving}
STATE	different aspects of the driver's physical state	{drug_level, alcohol_level, fatigue}

Table 2 Sorts used

To formalise the dynamic relationships between these concepts, the following relationships are used, expressed in LEADSTO:

DDR1 (Consuming a substance)

If a person consumes a certain product (e.g., drugs),
 and this product affects a certain aspect of the driver's state (e.g., drug level in his blood),
 then this aspect will be high (e.g., (s)he will have a high concentration of drugs in the blood)

consumed(P) & affects(P,S)
 $\rightarrow$ has_level(S, high)

DDR2 (Not consuming a substance)

If a person does not consume a certain product,
 and this product affects a certain aspect of the driver's state,
 then this aspect will be low

not(consumed(P)) & affects(P,S)
 $\rightarrow$ has_level(S, low)

Note that relationship DDR1 and DDR2 can also be replaced by mathematical formulae, stating, for example, how the consumption of alcohol influences the concentration of alcohol in the blood, and how this concentration decreases over time. To this end, more detailed medical knowledge (e.g., about drug and alcohol uptake by the blood) could be exploited.

DDR3 (From high drug level to abnormal steering)

If a person has a high drug level,
 then (s)he will perform abnormal steering behaviour

has_level(drug_level, high)
 → performed(abnormal_steering_behaviour)

DDR4 (From high alcohol level to abnormal steering and staring)

If a person has a high alcohol level,
 then (s)he will perform abnormal steering behaviour
 and (s)he will stare

has_level(alcohol_level, high)
 → performed(abnormal_steering_behaviour) $\wedge$ performed(staring)

DDR5 (From fatigue to staring)

If a person has a high state of fatigue,
 then (s)he will stare

has_level(fatigue, high)
 → performed(staring)

Finally, a number of relationships are used to model the processes related to eye blinking, as shown below. Here, d1 and d2 are constants that represent some basic eye dryness in case of, respectively, a high or low state of fatigue. Moreover, s1 and s2 represent the speed by which the eyes become drier in case of, respectively, a high or low state of fatigue.

DDR6 (Effect of fatigue on eye dryness)

If a person has a high state of fatigue,
 then the person's basic eye dryness will be d1,
 and the person's eye drying speed will be s1,

has_level(fatigue, high)
 → basic_eye_dryness(d1) & eye_drying_speed(s1)

DDR7 (Basic eye dryness)

If a person has a low state of fatigue,
 then the person's basic eye dryness will be d2,
 and the person's eye drying speed will be s2,

has_level(fatigue, low)
 → basic_eye_dryness(d2) & eye_drying_speed(s2)

DDR8 (Eye blinking)

If a person's eye dryness is X,
 and the person's eye dryness norm is Y,
 and $X > Y$,
 then the person will perform an eye blink

eye_dryness(X) &
 eye_dryness_norm(Y) &
 $X > Y$
 → performed(eye_blink)

DDR9 (Effect of blinking on dryness)

If a person performs an eye blink,
 and the person's basic eye dryness is d ,
 then the person's eye dryness will be d

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performed(eye_blink) &
basic_eye_dryness(D)
→ eye_dryness(D)

```

DDR10 (Effect of not blinking on dryness)

If a person does not perform an eye blink,
 and the person's eye dryness is X ,
 and the person's eye drying speed is S ,
 then the person's eye dryness will be $X+S*(1-X)$

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not(performed(eye_blink)) &
eye_dryness(X) &
eye_drying_speed(S)
→ eye_dryness(X+S*(1-X))

```

2.3 Example Simulation

This section presents an example simulation trace that was generated on the basis of the domain model for driver behaviour. This trace is displayed in Figure 2. The settings for this trace were: $d1=0.1$, $d2=0.2$, $s1=0.05$, $s2=0.15$, and $\text{eye_dryness_norm}=0.5$. The upper graph denotes the symbolic predicates, and the lower to graphs denote the numerical concepts (i.e., eye dryness). Note that predicates of type `affects(...)` have been left out, for readability.

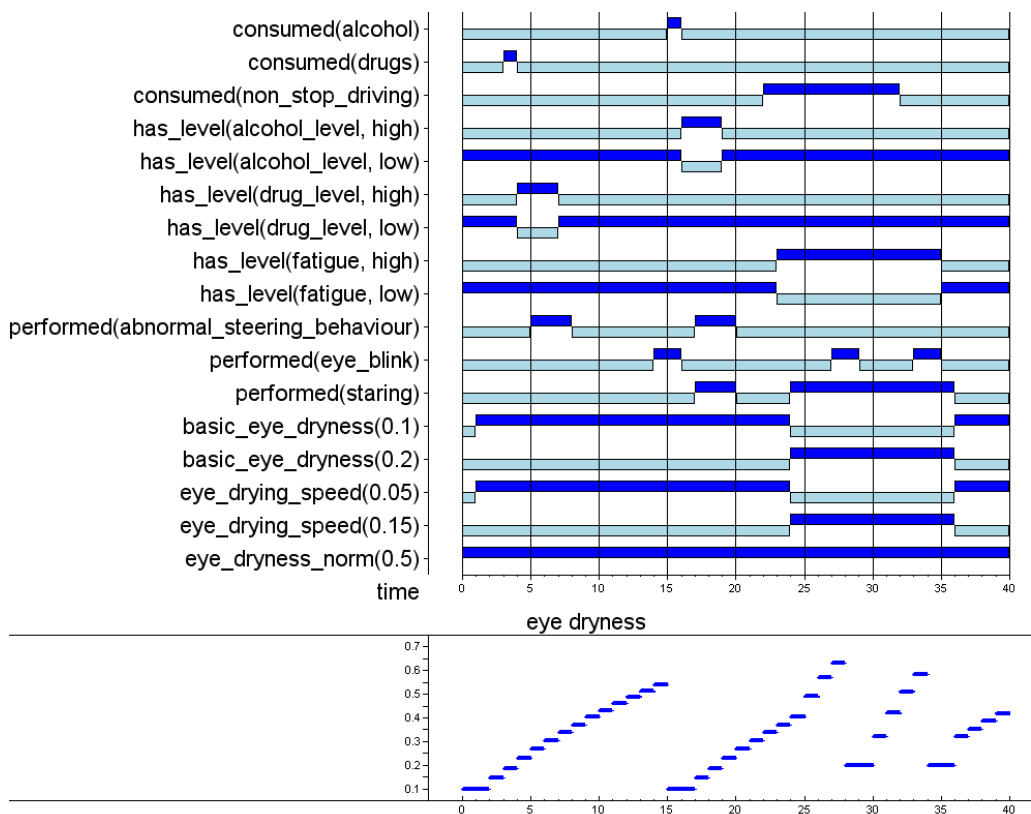


Figure 2 Example simulation trace for domain model.

As shown in Figure 2, the person involved initially has a low concentration of drugs and alcohol in the blood, and a low state of fatigue. However, at time point 3 (s)he uses some drugs, which leads to an increased drug level between time point 4 and 7 (it is assumed that this lasts 3 time steps), and abnormal steering behaviour. After that, everything is back to normal, until time point 15. At that moment, the person consumes some alcohol, which leads to an increased alcohol level for a while. As a result of this, the person -again- steers abnormally, and now also stares in the distance. However, until that moment, the person has not been very tired (i.e., a low state of fatigue), which resulted in a basic eye dryness of 0.1 and an eye drying rate of 0.05. As a consequence, the person's eyes have been drying very slowly, and (s)he has blinked only once (at time point 14, when the eyes were drier than the norm of 0.5). However, between time point 22 and 32, the person enters a long period of non-stop driving. As a result, his state of fatigue becomes high, and his values for basic eye dryness and eye drying rate increase. This causes the eyes to dry rather quickly, so that the person blinks twice in a relatively short period. After time point 36, everything is back to normal again.

3 Model for Analysis

This section introduces the model for analysis of driver behaviour. In Section 3.1, the main aspects of the model and their relations are introduced. In Section 3.2, the detailed model is provided, and Section 3.3 presents an example simulation trace.

3.1 Analysis of the Main Aspects and Their Relations

To be able to analyse the dynamics of a driver's behaviour, an ambient agent should, in principle, be equipped with the domain model introduced in Section 2, and should be able to reason on the basis of this model. Therefore, the model for analysis should contain more or less the same concepts as the domain model. Below, an analysis model to assess the driver's state of fatigue is worked out in detail. This means that the parts of the domain model that address fatigue are re-used here; to keep the model simple, the parts about drug and alcohol intake are not addressed.

The input that the ambient agent can use to make estimations about the state of fatigue is the amount of *non-stop driving*. The agent can *observe* this, because it knows when the engine is turned on and off. Based on that information, it can derive *beliefs* about the driver's state of *fatigue*. In combination with the agent's *desires* about the driver's state of *fatigue*, this can be used to determine whether there is a *discrepancy* between the driver's believed and desired state of fatigue. However, this is not the only useful information the agent can exploit. Because of the camera on the front mirror, it can also *observe* the driver's *eye blinks*. Therefore, it is useful for the agent to also make predictions (modelled as *beliefs*) about when

the driver will perform these *eye blinks*, so that it can compare this with the observed eye blinks in order to update its knowledge in case its beliefs (that were generated only on the basis of the information about non-stop driving) are not entirely correct. Possible mechanisms to make such updates are discussed in Section 5. In order to make predictions about eye blinks, the agent may use several of the relationships of the domain model (modelled in the form of *beliefs*). These beliefs may involve *eye dryness*, *basic eye dryness*, *eye drying speed*, and *eye dryness norm*.

This leads to the following list of concepts that are needed for the driver behaviour analysis model:

- observations on non-stop driving
- observations on eye blinks
- beliefs on fatigue
- beliefs on eye blinks
- beliefs on eye dryness
- beliefs on basic eye dryness
- beliefs on eye drying speed
- beliefs on eye dryness norm
- desires on fatigue
- beliefs on fatigue discrepancy

Based on these concepts, the following dynamic relationships are formulated:

Affecting beliefs on fatigue

- observations on non-stop driving affects beliefs on fatigue

Affecting beliefs on basic eye dryness and drying speed

- beliefs on fatigue affect beliefs on basic eye dryness
- beliefs on fatigue affect beliefs on eye drying speed

Affecting beliefs on eye dryness

- observations on eye blinks affect beliefs on eye dryness
- beliefs on basic eye dryness affect beliefs on eye dryness
- beliefs on eye drying speed affect beliefs on eye dryness

Affecting beliefs on eye blinks

- beliefs on eye dryness affect beliefs on eye blinks
- beliefs on eye dryness norm affect beliefs on eye blinks

Affecting beliefs on fatigue discrepancy

- desires on fatigue affect beliefs on fatigue discrepancy
- beliefs on fatigue affect beliefs on fatigue discrepancy

These dynamic relationships are visualised in Figure 3.

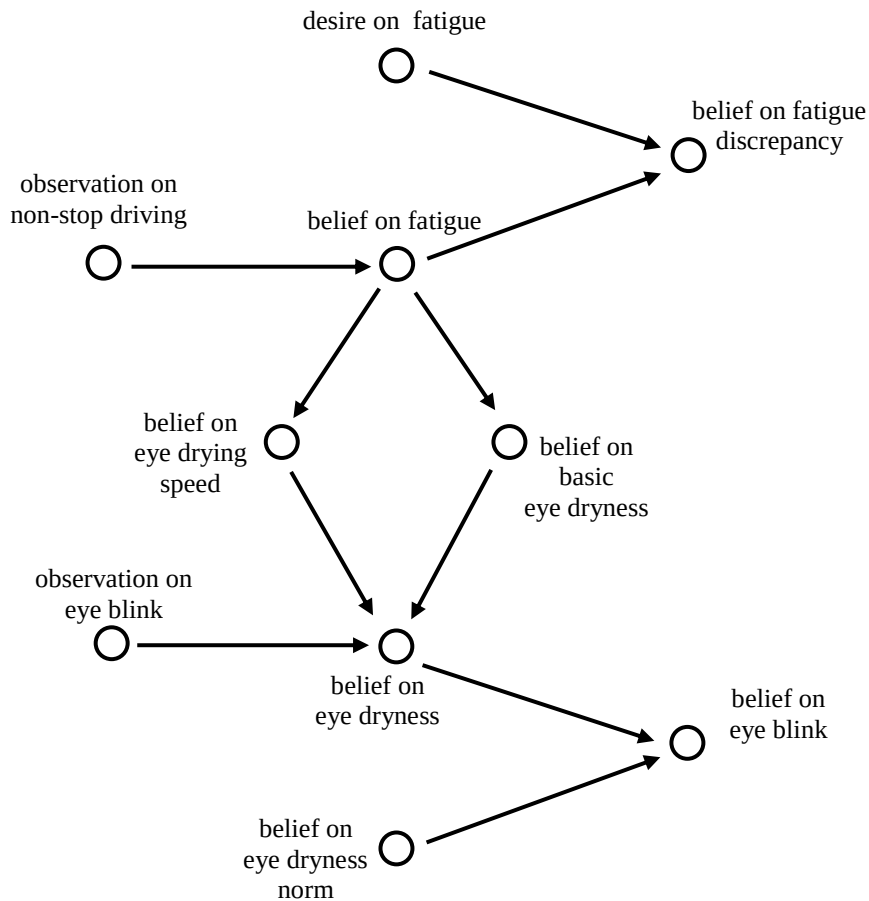


Figure 3 Overview of Analysis Model for Driver Behaviour

3.2 The Detailed Model

To formalise the concepts introduced in the previous section, again a number of logical atoms are introduced; see Table 3.

<i>concept</i>	<i>formalisation</i>
observations on non-stop driving	<code>observed(agent, consumed(non_stop_driving))</code>
observations on eye blinks	<code>observed(agent, performed(eye_blink))</code>
beliefs on fatigue	<code>belief(agent, has_level(fatigue, L:LEVEL))</code>
beliefs on eye blinks	<code>belief(agent, performed(eye_blink))</code>
beliefs on eye dryness	<code>belief(agent, eye_dryness(I:REAL))</code>
beliefs on basic eye dryness	<code>belief(agent, basic_eye_dryness(I:REAL))</code>
beliefs on eye drying speed	<code>belief(agent, eye_drying_speed(I:REAL))</code>
beliefs on eye dryness norm	<code>belief(agent, eye_dryness_norm(I:REAL))</code>
desires on fatigue	<code>desire(agent, has_level(fatigue, L:LEVEL))</code>
beliefs on fatigue discrepancy	<code>belief(agent, has_discrepancy(fatigue, L:LEVEL))</code>
beliefs on relations between factors and driver states	<code>belief(agent, affects(P:PRODUCT, S:STATE))</code>

Table 3 Formalisation of concepts used for analysis model

Next, to formalise the dynamic relationships for the analysis model, the following relationships in LEADSTO are used:

ADR1 (Generating beliefs for consuming a substance)

if the agent observes that a person consumes a certain product (e.g., non-stop driving),
 and the agent believes that this product affects a certain aspect of the driver's state (e.g., fatigue),
 then the agent will believe that this aspect will be high

$$\text{observed}(\text{agent}, \text{consumed}(P)) \wedge \text{belief}(\text{agent}, \text{affects}(P, S))$$

$$\rightarrow \text{belief}(\text{agent}, \text{has_level}(S, \text{high}))$$

ADR2 (Generating beliefs for not consuming a substance)

if the agent observes that a person does not consume a certain product,
 and the agent believes that this product affects a certain aspect of the driver's state,
 then the agent will believe that this aspect will be low

$$\text{not}(\text{observed}(\text{agent}, \text{consumed}(P))) \wedge \text{belief}(\text{agent}, \text{affects}(P, S))$$

$$\rightarrow \text{belief}(\text{agent}, \text{has_level}(S, \text{low}))$$

ADR3 (Generating beliefs for effect of high fatigue on eye dryness)

if the agent believes that a person has a high state of fatigue,
 then the agent believes that the person's basic eye dryness will be d1,
 and the agent will believe that the person's eye drying speed will be s1,

$$\text{belief}(\text{agent}, \text{has_level}(\text{fatigue}, \text{high}))$$

$$\rightarrow \text{belief}(\text{agent}, \text{basic_eye_dryness}(d1)) \wedge \text{belief}(\text{agent}, \text{eye_drying_speed}(s1))$$

ADR4 (Generating beliefs for effect of low fatigue on eye dryness)

if the agent believes that a person has a low state of fatigue,
 then the agent believes that the person's basic eye dryness will be d2,
 and the agent will believe that the person's eye drying speed will be s2,

$$\text{belief}(\text{agent}, \text{has_level}(\text{fatigue}, \text{low}))$$

$$\rightarrow \text{belief}(\text{agent}, \text{basic_eye_dryness}(d2)) \wedge \text{belief}(\text{agent}, \text{eye_drying_speed}(s2))$$

ADR5 (Generating beliefs for eye blinking)

if the agent believes that a person's eye dryness is X,
 and the agent believes that the person's eye dryness norm is Y,
 and $X > Y$,
 then the agent will believe (predicts) that the person will perform an eye blink

$$\text{belief}(\text{agent}, \text{eye_dryness}(X)) \wedge \text{belief}(\text{agent}, \text{eye_dryness_norm}(Y)) \wedge x > y$$

$$\rightarrow \text{belief}(\text{agent}, \text{performed}(\text{eye_blink}))$$

ADR6 (Generating beliefs for eye dryness)

if the agent observes that a person performs an eye blink,
 and the agent believes that the person's basic eye dryness is d,
 then the agent will believe that the person's eye dryness will be d

$$\text{observed}(\text{agent}, \text{performed}(\text{eye_blink})) \wedge \text{belief}(\text{agent}, \text{basic_eye_dryness}(D))$$

$$\rightarrow \text{belief}(\text{agent}, \text{eye_dryness}(D))$$

ADR7 (Generating beliefs for effect of not blinking on dryness)

if the agent observes that a person does not perform an eye blink,
 and the agent believes that the person's eye dryness is X,
 and the agent believes that the person's eye drying speed is S,
 then the agent will believe that the person's eye dryness will be $X+S*(1-X)$
 $\text{not}(\text{observed}(\text{agent}, \text{performed}(\text{eye_blink}))) \wedge \text{belief}(\text{agent}, \text{eye_dryness}(X)) \wedge \text{belief}(\text{agent}, \text{eye_drying_speed}(S))$
 $\rightarrow \text{belief}(\text{agent}, \text{eye_dryness}(X+S*(1-X)))$

ADR8 (Assessing fatigue)

if the agent desires that the driver has a low state of fatigue,
 and the agent believes that the driver has a high state of fatigue,
 then the agent will believe that there is a (high) discrepancy
 between the driver's desired and believed state of fatigue
 $\text{desire}(\text{agent}, \text{has_level}(\text{fatigue}, \text{low})) \wedge \text{belief}(\text{agent}, \text{has_level}(\text{fatigue}, \text{high}))$
 $\rightarrow \text{belief}(\text{agent}, \text{has_discrepancy}(\text{fatigue}, \text{high}))$

3.3 Example Simulation

Also for the analysis model, simulation runs may be performed. An example simulation trace for a situation in which the ambient agent signals a discrepancy is shown in Figure 4.

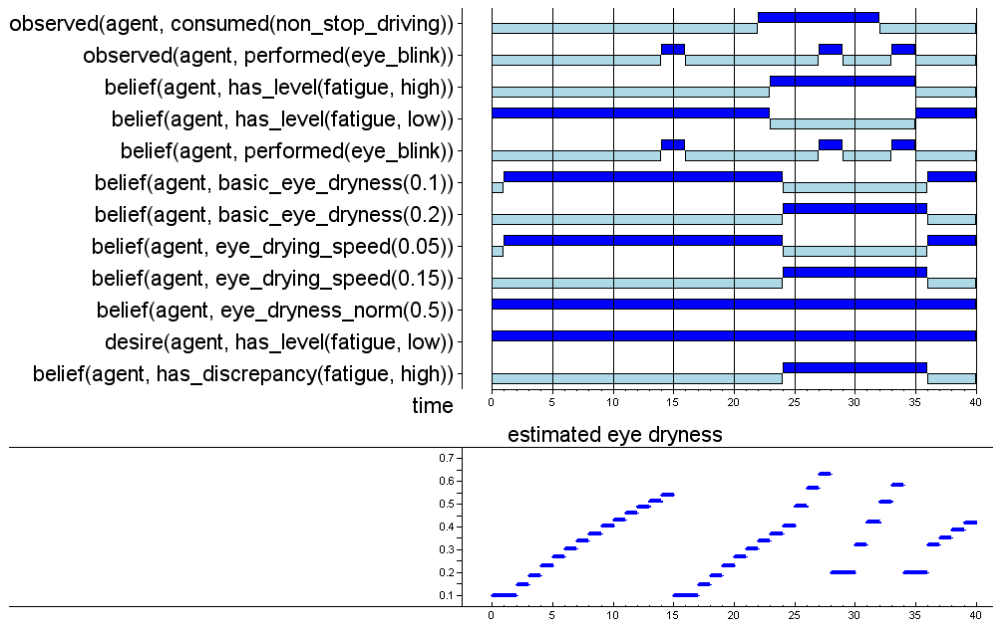


Figure 4 Example simulation trace for analysis model.

As shown in the figure, this scenario is quite similar to the scenario shown in Figure 2 (with all information related to drug and alcohol intake left out). The driver starts with a low state of fatigue, which means that (s)he performs eye blinks at a low frequency. However, between time point 22 and 32, the person drives a long distance without taking a break, which is observed by the agent. Based on this, the agent believes that (s)he is in a high state of fatigue. Since this is in conflict with the agent's desire (namely that the driver is in a low state of

fatigue), the agent derives that there is a fatigue discrepancy (which can be input for the support model introduced in the next section).

Furthermore, notice that, based on its belief that the driver is in a high state of fatigue, the agent also predicts that the driver's eyes will become dry very quickly, and that (s)he will blink at time point 27 and 33. Since the agent's analysis model is 100% correct, these predictions correspond exactly to reality. However, in cases that the analysis model is not that perfect, the predicted eye blinks will not exactly overlap with the observed eye blinks. In such cases, the parameters used by the model will have to be updated. This can be done, for example, using mechanisms as described in Section 5.

Note that the model presented in this section can only be used to assess fatigue based on observations about non-stop driving (and eye blinks). Thus, it covers the bottom part of Figure 1. However, the upper part of that picture has not been worked out here. To this end, an analysis model to assess the driver's drug and alcohol concentration should be developed. One additional difficulty in that case would be that some relevant concepts that exist in the domain model cannot directly be observed by the ambient agent. For example, the amount of alcohol and drug intake cannot easily be measured by an ambient agent. On the other hand, it is possible to measure information about whether the driver shows abnormal steering behaviour or is staring (the concepts on the upper-right hand side of Figure 1). These types of information can be measured, respectively, by the movements of the steering wheel and the camera on the front mirror.

Thus, it is still possible to obtain information about states in these domain models, but these state are only present at the right-hand side of the model. As a result, no simple forward reasoning pattern (of the format "if I believe X and I believe that X leads to Y, then I also believe Y") can be applied. Instead, other reasoning methods should be applied. An example of such a method is *abduction*. In a nutshell, this method has the format "if I believe Y and I believe that X leads to Y, then I also believe that X could have been the case". By applying such methods, one could derive, for example, that the driver has been drinking on the basis of an abnormal steering pattern that is observed (e.g., in case the steering wheel moves more than 3 times from left to right and back within five seconds). For more information about such methods, for example, see Chapter 2.

In case different analysis models are developed, which each analyse one particular aspect of the driver's state (e.g., drug concentration, alcohol concentration, and state of fatigue), the results of analyses can also be combined, in order to provide more evidence for a certain hypothesis (e.g., in case the driver has a high blinking frequency and has not stopped the car for a while, (s)he is very likely to have a high state of fatigue). To this end, it may be useful to

develop a multi-agent system, consisting of different (monitoring) agents that each analyse a specific factor separately, and another agent that combines the conclusions of the monitoring agents. Such an approach is taken in (Bosse, Hoogendoorn, Klein, and Treur, 2008). An overview of the multi-agent system used in that paper is shown in Figure 5.

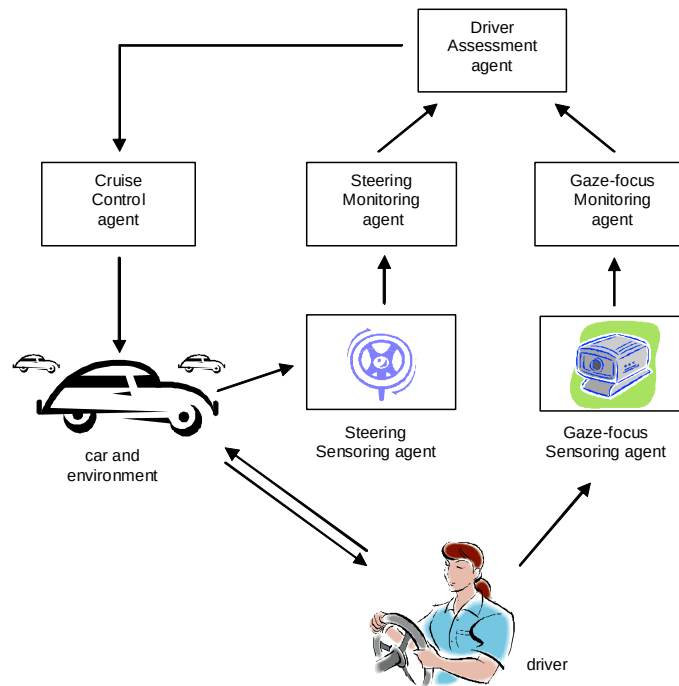


Figure 5 Multi-agent system for ambient driver support (Bosse, Hoogendoorn, Klein, and Treur, 2008).

Here, the Steering Monitoring agent receives the sensor input from the Steering Sensing agent, and uses this to determine whether there is an abnormal steering pattern. Similarly, the Gaze-focus Monitoring agent uses sensor input from the Gaze-focus Sensing agent to determine whether there is an abnormal gaze-focus pattern. Next, the Driver Assessment agent combines the information from the two monitoring agents to determine whether the driver is in an impaired condition. If this is the case, the Driver Assessment agent sends a signal to the Cruise Control agent, who can support the driver, e.g., by slowing down the car. Development of such a system composed of multiple cooperating agents is left as an exercise for the reader.

4 Model for Support

This section introduces the model for support of a car driver. In Section 4.1, the main aspects of the model and their relations are introduced. In Section 4.2, the detailed model is provided, and Section 4.3 presents an example simulation trace.

4.1 Analysis of the Main Aspects and Their Relations

In case an ambient agent in a car signals that the driver is in an impaired condition, it has several alternatives to provide support. Some possible support actions that are mentioned in the literature are:

- keep a safe distance with the car in front
- slow down the car and safely pull it over
- wake up a sleepy driver (e.g., by making a loud sound)
- give advice to the driver (e.g., tell him or her to take a break)

The choice for which action to perform may depend on the specific state the driver is in. For example, if the agent believes that the driver is only slightly tired, it may advice him or her to take a break. However, when it estimates that the driver is practically asleep, a more urgent measure is necessary.

In sum, the following concepts are needed for the driver support model:

- beliefs about discrepancies between the driver's believed and desired drug level
- beliefs about discrepancies between the driver's believed and desired alcohol
- beliefs about discrepancies between the driver's believed and desired state of fatigue
- actions to keep distance with the car in front
- actions to slow down the car
- actions to wake up a sleepy driver
- actions to give advice to the driver

For the dynamic relationships between these concepts, multiple options are possible that connect a specific belief (or a combination of beliefs) to a particular action. In the next section, an example specification is proposed. The relationships used in that example are visualised in Figure 6.

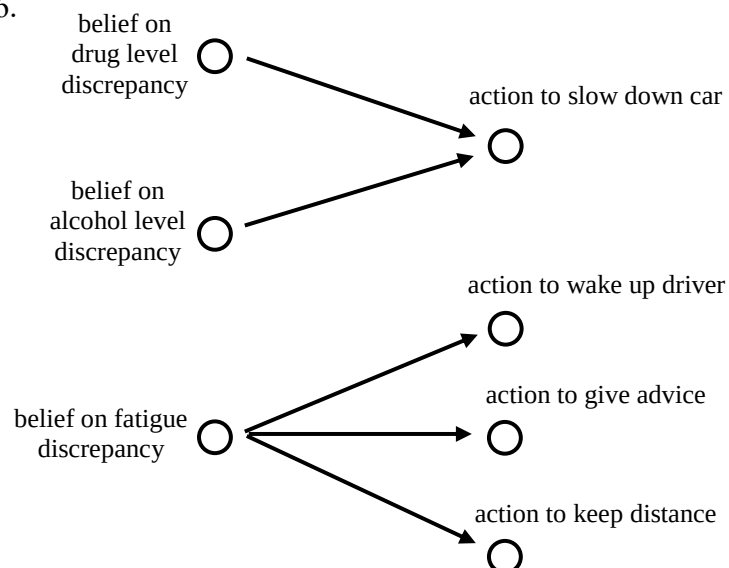


Figure 6 Overview of Support Model for Driver Behaviour

4.2 The Detailed Model

To formalise the concepts introduced in the previous section, the following logical atoms are used; see Table 4.

<i>concept</i>	<i>formalisation</i>
beliefs of discrepancies of the driver's drug level	belief(agent, has_discrepancy(drug_level, L:LEVEL))
beliefs of discrepancies of the driver's alcohol level	belief(agent, has_discrepancy(alcohol, L:LEVEL))
beliefs of discrepancies of the driver's state of fatigue	belief(agent, has_discrepancy(fatigue, L:LEVEL))
actions to keep distance with the car in front	performed(agent, keep_distance)
actions to slow down the car	performed(agent, slow_down_car)
actions to wake up a sleepy driver	performed(agent, wake_up_driver)
actions to give advice	performed(agent, give_advice)

Table 4 Formalisation of concepts used for support model

To formalise dynamic relationships that connect beliefs to support actions, the following relationships in LEADSTO are used:

SDR1 (Generating a wake up action)

if the agent believes that there is a high discrepancy with respect to the driver's state of fatigue,
 then the agent will try to wake up the driver
 $\text{belief}(\text{agent}, \text{has_discrepancy}(\text{fatigue}, \text{high}))$
 $\rightarrow \text{performed}(\text{agent}, \text{wake_up_driver})$

SDR2 (Generating advice to keep distance)

if the agent believes that there is a low discrepancy with respect to the driver's state of fatigue,
 then the agent will give the driver advice to take a break
 $\text{belief}(\text{agent}, \text{has_discrepancy}(\text{fatigue}, \text{low}))$
 $\rightarrow \text{performed}(\text{agent}, \text{give_advice}) \wedge \text{performed}(\text{agent}, \text{keep_distance})$

SDR3 (For alcohol slowing down the car)

if the agent believes that there is a high discrepancy with respect to the driver's alcohol level,
 then the agent will try to slow down the car
 $\text{belief}(\text{agent}, \text{has_discrepancy}(\text{alcohol_level}, \text{high}))$
 $\rightarrow \text{performed}(\text{agent}, \text{slow_down_car})$

SDR4 (For drugs slowing down the car)

if the agent believes that there is a high discrepancy with respect to the driver's drug level,
 then the agent will try to slow down the car
 $\text{belief}(\text{agent}, \text{has_discrepancy}(\text{drug_level}, \text{high}))$
 $\rightarrow \text{performed}(\text{agent}, \text{slow_down_car})$

Note that this knowledge base takes a relatively simple approach, based on a 1:1 mapping between an estimated driver state and a set of support actions. The reader is encouraged to develop other, more sophisticated methods for support. For example, the agent could first

generate a number of support options, and then select the one with the highest feasibility (see, for instance, Chapter 6).

4.3 Example Simulation

Simulation of the model presented in Section 4.2 is straightforward, and therefore not worked out here.

5 Model for Parameter Adaptation

This section introduces the model for adaptation of the parameters of the driver behaviour model. In Section 5.1, the main aspects of the model and their relations are introduced. In Section 5.2, the detailed model is provided, and Section 5.3 presents an example simulation trace.

5.1 Analysis of the Main Aspects and Their Relations

The parameters that the agent uses in its analysis (and support) model need not exactly correspond to the actual parameters that describe reality. When parameter values used by the ambient agent differ too much from reality, the agent may make errors in its analysis and in its support actions. Therefore it is better to design the agent in such a way that it can learn from these errors, and after discovering them, adapts its parameter values. This section presents a model for such an adaptation process. The model uses the same mechanism for parameter adjustment as presented in Chapter 6 (see that chapter for more details).

As in Chapter 6, two main questions to be addressed are (1) how does the agent get information on (a certain extent of) inaccuracy or incorrectness of the model used, and (2) how can it relate such information to deviations in parameter values that have to be compensated. For the current domain, an answer to the first question would be “by calculating the difference between observed and predicted eye blink frequency”, or “by calculating the difference between observed and predicted alcohol concentration”. To keep the example simple, in this section the focus is on the driver’s level of alcohol in the blood only (so drug level and state of fatigue are not considered, although they can be addressed using similar mechanisms). It is assumed that from time to time *observations about alcohol level* are provided via a sensor in the steering wheel that measures the alcohol concentration in the sweat. By comparing this with the *estimated level of alcohol* (based on mechanisms as described in Section 3), the agent can determine what is the inaccuracy in its model (in terms of *deviations*). Note that this also assumes that alcohol concentration is not represented on a discrete scale (as done in the previous sections, using values low and high only), but on a continuous scale, via a real number. Concerning the second question, in this chapter it is assumed that the *parameter* in the model that has to be compensated is known. Since the model to estimate alcohol concentration is not worked out in Section 3, it is not possible to

provide a concrete example for such a parameter³. However, below the generic case is described, for any parameter P . To discover deviations in such parameters, the presented approach makes use of *sensitivities*, *adaptation speeds*, and *adaptation options*, similar to Chapter 6.

Thus, the following concepts are needed for the driver behaviour parameter adaptation model:

- beliefs about the driver's alcohol level in the blood
- observations about the driver's alcohol level in the blood
- beliefs about deviations between the estimated and observed alcohol level
- parameter values
- sensitivities
- adaptation speeds
- adaptation options

The following dynamic relationship between these concepts can be established:

- beliefs about the driver's alcohol level in the blood affect beliefs about deviations
- observations about the driver's alcohol level in the blood affect beliefs about deviations
- beliefs about deviations affect adaptation options
- sensitivities affect adaptation options
- adaptation speeds affect adaptation options
- adaptation options affect parameter values

These dynamic relationships are visualised in Figure 7.

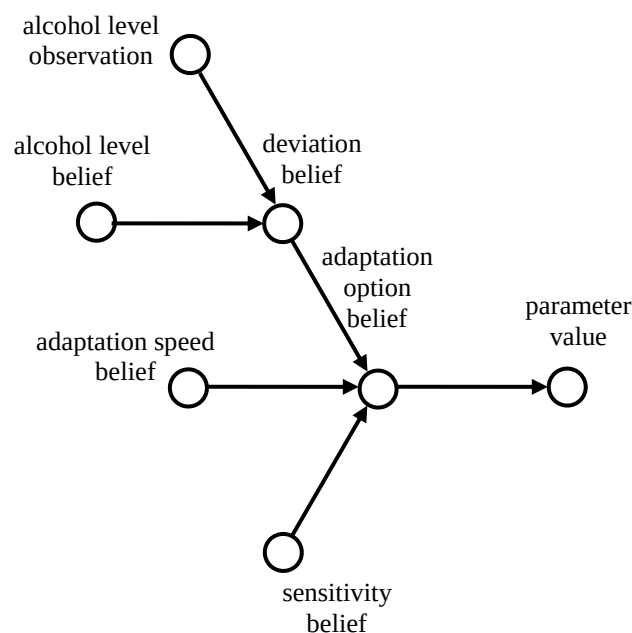


Figure 7 Overview of Parameter Adaptation Model for Driver Behaviour

³ However, for the fatigue case, concrete examples can be provided. For example, the eye drying speed, the basic eye dryness, or the drying norm.

5.2 The Detailed Model

The following logical atoms are used to formalise the concepts introduced above; see Table 5. Note that, since the change of alcohol concentration in the blood is a highly dynamic process, it is needed to represent explicitly at which time point a certain concentration is observed or estimated. To this end, a parameter t for time is included in some of the atoms.

<i>concept</i>	<i>formalisation</i>
beliefs about the driver's alcohol level in the blood at time t	$\text{belief}(\text{agent}, \text{has_level_at}(\text{alcohol_level}, W:\text{REAL}, T:\text{REAL}))$
observations about the driver's alcohol level in the blood	$\text{observed}(\text{agent}, \text{has_level_at}(\text{alcohol_level}, W:\text{REAL}, T:\text{REAL}))$
beliefs about deviations between the estimated and observed alcohol level	$\text{belief}(\text{agent}, \text{has_deviation}(\text{has_level_at}(\text{alcohol_level}, W1:\text{REAL}, T:\text{REAL}), W2:\text{REAL}))$
beliefs on parameter values	$\text{belief}(\text{agent}, \text{has_value}(P:\text{PARAMETER}, V:\text{REAL}))$
sensitivities	$\text{belief}(\text{agent}, \text{has_sensitivity_for}(\text{has_level_at}(\text{alcohol_level}, W:\text{REAL}, T:\text{REAL}), S:\text{REAL}, P:\text{PARAMETER}, V:\text{REAL}))$
adaptation speeds	$\text{belief}(\text{agent}, \text{adaptation_speed}(U:\text{REAL}))$
adaptation options	$\text{belief}(\text{agent}, \text{adaptation_option}(\text{adapt_to}(P:\text{PARAMETER}, V1:\text{REAL}, V2:\text{REAL})))$

Table 5 Formalisation of concepts used for parameter adaptation model

Here, the sort `PARAMETER` contains all parameters that occur in the model. Moreover, note that the element `alcohol_level` can be replaced by `S:STATE` for the more generic case.

PADR1 (Generating deviation beliefs)

If the agent believes that the level of alcohol at T is $W1$
 and the agent observes that the level of alcohol at T is $W2$
 then the agent will believe that the level of alcohol at T has a deviation of $W2-W1$

$\text{belief}(\text{agent}, \text{has_level_at}(\text{alcohol_level}, W1, T)) \ \& \$
 $\text{observed}(\text{agent}, \text{has_level_at}(\text{alcohol_level}, W2, T))$
 $\rightarrow \text{belief}(\text{agent}, \text{has_deviation}(\text{has_level_at}(\text{alcohol_level}, W1, T), W2-W1))$

PADR2 (Generating beliefs on adaptation options 1)

If the agent believes that level of alcohol W at T has sensitivity S for parameter P with value V
 and that a deviation D for level of alcohol W at T was identified
 and $D/S \geq 0$
 and the agent believes that the adaptation speed is U
 then the agent will believe that an option is to change the value V for P to $V + U \cdot D \cdot (1-V)/S$

$\text{belief}(\text{agent}, \text{has_sensitivity_for}(\text{has_level_at}(\text{alcohol_level}, W, T), S, P, V)) \ \& \$
 $\text{belief}(\text{agent}, \text{has_deviation}(\text{has_level_at}(\text{alcohol_level}, W, T), D)) \ \& \$
 $D/S \geq 0 \ \& \$
 $\text{belief}(\text{agent}, \text{adaptation_speed}(U))$
 $\rightarrow \text{belief}(\text{agent}, \text{adaptation_option}(\text{adapt_to}(P, V, V+U \cdot D \cdot (1-V)/S)))$

PADR3 (Generating beliefs on adaptation options 2)

If the agent believes that level of alcohol W at T has sensitivity S for parameter P with value V
 and that a deviation D for level of alcohol W at T was identified
 and $D/S \leq 0$
 and the agent believes that the adaptation speed is U

then the agent will believe that an option is to change the value V for P to $V - U \cdot D \cdot V / S$

```

belief(agent, has_sensitivity_for(has_level_at(alcohol_level, W, T), S, P, V) &
belief(agent, has_deviation(has_level_at(alcohol_level, W, T), D)) &
D/S ≤ 0 &
belief(agent, adaptation_speed(U))
→ belief(agent, adaptation_option(adapt_to(P, V, V-U*D*V/S)))

```

PADR4 (Adapting parameter beliefs)

If the agent believes that an adaptation option is to change the value $V1$ for parameter P to $V2$

then the agent will believe that the value of parameter P is $V2$

and it will not believe that the value of parameter P is $V1$

```

belief(agent, adaptation_option(adapt_to(P, V1, V2)))
→ belief(agent, has_value(P, V2)) & not belief(agent, has_value(P, V1))

```

Note that the question how sensitivities can be determined is not addressed here. See Chapter 6 for a detailed description about this.

5.3 Example Simulation

An example simulation trace for the parameter adaptation model described above is shown in Figure 8. As shown in this figure, the agent initially believes that the alcohol concentration in the driver's blood is 0.05%. This is based on an extensive analysis process (which is not depicted in Figure 8, which involves the value of 0.7 for parameter P). However, at time point 7 the agent receives more precise information about the driver's alcohol level via the sensor in the steering wheel: the agent observes that the actual alcohol concentration in the driver's blood is 0.08. As a result, the agent notices that there is a deviation of 0.03, and decides to adjust the value of parameter P to 0.679.

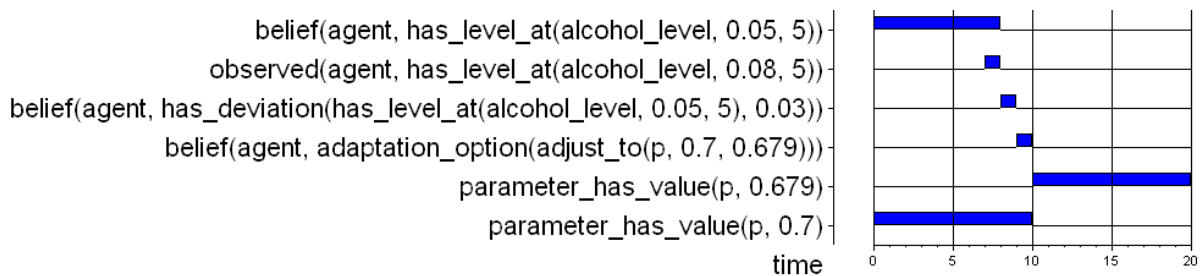


Figure 8 Example simulation trace for parameter adaptation model.

Note that the values for the sensitivity S for parameter P , and the adaptation speed U are not shown in Figure 8. In the current scenario, both have been set by hand to 0.1. However, in a more realistic scenario, these values could be dynamic.

6 Discussion

In order to assess whether a driver is capable of driving a car, an ambient agent needs to maintain a domain model of the state of alertness this person. This chapter introduced such a

domain model (in Section 2), which involved various risk factors such as alcohol intake, drug intake, and periods of non-stop driving. In addition, an analysis model was introduced to reason about this domain model (Section 3), as well as a support model to determine actions based on the analysis (Section 4), and a model to adapt the parameter values of the analysis model (Section 5).

The presented model can be extended in various manners. Some possibilities are mentioned below:

- Taking more factors into account than alcohol, drugs, and fatigue. For example, emotions, other passengers, music, or mobile phones.
- Modelling the concentration of alcohol and drugs in more detail, using a continuous, numerical approach.
- Developing more complex models for analysis (e.g., a model to analyse steering patterns by taking speed and direction of steering movements into account).
- Designing a multi-agent system (instead of a single agent system) for analysis and support of driver behaviour.
- Considering different and more complex types of support (e.g., a mechanism to compare different options and select the one with the highest feasibility).
- Considering more sophisticated techniques for adaptation of parameter values.

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